

Blen Ermias

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EDUCATION

Portland State University | Portland, OR

Expected: **June 2028**

B.S. Electrical Engineering, Maseeh College of Engineering and Computer Science | Honors College

TECHNICAL EXPERIENCE

Portland State Aerospace Society (PSAS) | Portland, OR

Firmware Research Intern

June 2026- Present

- Researching low-power optimization techniques for embedded systems using the Zephyr Real-Time Operating System (RTOS) on the OreSat Protocard, a card featuring the NXP MCXN947 MCU and supporting hardware.
- Develop and test Zephyr RTOS firmware on the NXP MCXN947 to characterize power consumption across processor sleep states and optimize energy use on the OreSat Protocard.
- Evaluate Zephyr's power management framework by investigating MCU sleep states, peripheral power management, clock gating, and timer behavior to identify sources of power consumption and optimize sleep/wake transitions.
- Extending this research into an Honors Thesis focused on low-power embedded systems for aerospace applications, with plans for future technical publication.

Computer Engineering Research Intern

June- Sep 2025

- Debugging, testing, and writing applications on NXP MCXN947 microcontroller using Zephyr RTOS.
- Wrote hardware descriptions, board-specific configurations, and system configuration files to build and flash applications for the OreSat1 breakout board, part of the PSAS open-source CubeSat project.
- Learned PCB design by tracing protoboards using KiCAD software.
- Presented embedded system research at the PSU Undergraduate Research Symposium.

PROJECTS

FlatSat Breakout Board

Dec 2025- Feb 2026

- Designed a FlatSat breakout PCB that integrates satellite subsystems for software development, testing, and debugging as part of the Portland State Aerospace Society's OreSat1 CubeSat project.
- Developed the schematic and PCB layout using KiCad, ensuring proper electrical connections and system integration.

Programmable Flashlight

Jan 2026 – Mar 2026

- Designed a programmable multifunction flashlight sponsored by PSU's Electronics Prototyping Lab.
- Implemented adjustable brightness, selectable colors, strobe/stroboscope functionality, and Morse code output.
- Designed the system for future feature expansion while maintaining a compact and user-friendly form factor.

Programmable Weight Scale

Sep 2025 – Dec 2025

- Built a digital weight scale using an ESP8266 microcontroller, HX711 load-cell amplifier, load cells, and liquid-crystal display (LCD). Programmed the ESP8266 to process load-cell measurements and display weight in pounds.

INVOLVEMENT

National Society of Black Engineers (NSBE)

Vice-President

Present

- Co-founder of the Portland State University chapter of the National Society of Black Engineers (NSBE).
- Promoted chapter events and membership by distributing flyers and posters across campus and managing social media outreach.

PSU EAGLES Program | Portland, OR

Dec 2024- June 2026

Student Summer Research Scholar

- Selected as a PSU Undergraduate Summer Research Scholar through the EAGLES Program to conduct research with the Portland State Aerospace Society (PSAS).

TECHNICAL SKILLS

Languages: C, Python **Technologies:** Git, Zephyr RTOS, Linux, KiCAD